



PRODUCT REQUIREMENT DOCUMENT (PRD)

Color Prediction & Premium Dice Betting Platform

METADATA & PLATFORM SPECIFICATIONS

Date:	June 2026
Stack:	React (Vite), Tailwind CSS, Shadcn/ui, Node.js, Express, MongoDB
Author:	Technical Product & Architecture Division
Version:	v1.3.0 (Official Build)

Section 1: Executive Summary & Scope

This Product Requirement Document (PRD) defines the complete functional specifications, frontend layouts, and mathematical/financial rules governing the integrated gaming and wallet systems. The platform combines a dynamic retail accessories feed with real-time prediction-based modules to maximize user retention and engagement.

1.1 Product Vision & Core Loops

The application integrates an e-commerce catalog featuring accessories (e.g. AuraPods, Chronos Watches, Mechanical Keyboards) directly into the betting framework. This enables users to utilize their wallet balances to either purchase physical accessories or place stakes in lottery predictions. The goal is to provide a smooth, low-latency, mobile-first Web App experience that matches the performance of native iOS/Android applications.

1.2 Role-Based Workspace Clearance Matrix

To facilitate proper administration and operation, the user context dynamics adapt profiles based on three clearance tiers:

1. Standard User: Access to all games, personal transaction histories, linked payment portals, and direct withdrawals.
2. Admin: Strictly read-only access to system monitors, transaction registries, user logs, and draw history metrics. No write/override operations are permitted.
3. Super Admin: Comprehensive read/write management credentials. Enables updating product parameters, configuring carousel banners, issuing promotional vouchers, and overriding user wallet balances directly.

Section 2: Color Prediction Game Specifications

2.1 Decoupled Game Lobbies & Persistence

The Color Prediction game is decoupled into four concurrent timing lobbies: 30 Seconds, 1 Minute, 2 Minutes, and 5 Minutes. Lobby timers are managed atomically using React refs. Switching tabs does not clear, pause, or reset active countdown loops. When changing rooms, current draft bet inputs and glow outcome animations are flushed from UI states to keep the display clean.

2.2 Number-to-Color Remapping & Payout Rules

Winning numbers scale from 1 to 10. The remapping configuration is established as follows:

- Green (Emerald): Assigned to Even numbers 2, 4, 6, 8 (Pays 1.9x payout).
- Red (Rose): Assigned to Odd numbers 1, 3, 7, 9 (Pays 1.9x payout).
- Violet (Yallet): Assigned to numbers 5 and 10 (Pays 4.5x payout).

Note on split outcomes: Number 5 acts as a split Red-Violet number, so a round result of 5 triggers winning payouts for both Red and Violet bet targets. Number 10 is only Violet.

Number Betting: Correctly predicting the exact drawn number (1-10) pays a massive 8x payout.

2.3 Countdown timer card & 5-Second Betting Lock

The timer remaining is rendered as a large, premium dark-digital card block (bg-slate-900 text-emerald-400 font-mono text-2xl). When the clock reaches 5 seconds, an absolute visual modal locks overlay triggers over the color/number betting deck, disabling all buttons and preventing any wagers from being confirmed. It displays a large countdown (00:05 to 00:01) to build tension.

Section 3: Dice Game (Dice Pro) Specs

3.1 Interactive Range Slider Mechanics

Dice Pro features a custom range slider track where users select their target bounds. The slider background track displays colored zones dynamically: Red (#fee2e2) for losing ranges and Green (#dcfce7) for winning ranges, divided at the exact selected coordinate.

3.2 Mathematical Formulas & safety Clamps

The win probability and payout multiplier scale simultaneously based on target value (t):

- Roll Over Win Chance (P): $100 - t$
- Roll Under Win Chance (P): t
- Range (10) Win Chance (P): Fixed 10% (target start bounds from 0 to 90)
- Multiplier Formula (M): $M = 98 / \text{Win Probability}$ (rounded to 2 decimal places)
- Potential Profit = (Bet Amount * M) - Bet Amount

Safety Clamp: To prevent the 1.00x multiplier riskless exploit, targets in Over/Under modes are strictly locked between 4.90 and 95.10. The maximum win probability is restricted to 95.10%, ensuring the minimum multiplier is always 1.03x and the maximum multiplier reaches exactly 20.00x.

3.3 Roll Resolution Scrambler

When the timer reaches 0, the digital screen triggers a 1.5s scrambled roll animation where numeric values cycle rapidly. Once resolved, a colored pin indicator highlights the final rolled result coordinate directly on the slider track.

Section 4: Financial Engine & Wallet Rules

4.1 Withdrawal Fee Progression Equation

Processing fees are added on top of the requested payout and deducted from the wallet balance. Let x be the payout amount requested. The fee $f(x)$ is defined piecewise:

- For $x \leq 100$: $f(x) = x * 0.09$ (9% flat, Rs. 9 fee for Rs. 100 payout)
- For $100 < x \leq 1000$: $f(x) = 9 + (x - 100) * 0.03$ (adds 3% on top of the base Rs. 9 fee per Rs. 100 block)
- For $x > 1000$: $f(x) = x * 0.03$ (stabilizes to a flat 3% of the total amount requested)

4.2 Destination Constraints & Name Security Lock

1. Real Money Withdrawals: Users can withdraw their real cash balance (`realBalance`) at any time. Wagering alerts are purely informational and do not block withdrawals.
2. Locked Name Attribute Policy: When linking a bank or UPI destination, the cardholder `holderName` input is read-only and locked to `user.name`. This prevents users from withdrawing funds into third-party accounts.
3. Limits: Single transaction withdrawal limits are set at Min Rs. 100 and Max Rs. 5,000.

4.3 Bonus Balance & Wagering Conversion

- Winnings Routing: Winnings from wagers placed using bonus credits are routed to `bonusBalance` if `requiredWager > 0`.
- Auto-Conversion: The instant `requiredWager` reaches 0, the entire bonus balance converts into withdrawable `realBalance` immediately.

Section 5: Referrals & VIP Milestones

5.1 Referral System & Play Commissions

- Direct Referral Bonus: Users receive a flat Rs. 10 reward when their referee completes a valid first deposit of Rs. 100 or more.
- Rebate Play Commission: Users earn dynamic play commissions based on their VIP tier. Commission is only active and displayed for levels VIP-5 or above (0.5% commission rate at VIP-5, scaling to 2.0% at VIP-20). For levels VIP-1 to VIP-4, the commission rate is 0.0% and is hidden in the VIP club views.

Section 6: Role-Based Access Control (RBAC) Matrix

Module Domain	Standard User	Admin	Super Admin
Game Betting	Read / Write (Place Bets)	Read-Only View	Read-Only View
Wallet Payouts	Write (Request Payout)	Read-Only View	Write (Approve / Deny)
Wallet Deposits	Write (Request Deposit)	Read-Only View	Write (Credit Wallet)
Promo Banners	Read-Only View	Read-Only View	Read / Write (Configure Banners)
Product Feed	Read / Write (Checkout)	Read-Only View	Read / Write (Manage Stock)
VIP & Vouchers	Read / Claim Available	Read-Only View	Read / Write (Issue Vouchers)
System Settings	No Access	Read-Only View	Read / Write (Manage Settings)